RAG Learning

1.3 Glaciated Landscapes

Eduqas GCSE Geography (C/4) — Component 1

1. Glacial Periods

Ice age: A prolonged period in Earth's history when global temperatures fell significantly, causing large sheets of ice and glaciers to advance across the land surface. Ice ages are not continuous — they alternate between *glacial periods* (colder, ice advances) and *interglacials* (warmer, ice retreats).

Pleistocene Epoch: The geological time period lasting from approximately 2.6 million years ago to 11,700 years ago, characterised by repeated glaciations. During the last glacial maximum (c. 20,000 years ago), ice covered much of Britain as far south as the Bristol Channel and Thames Valley.

Little Ice Age: A cooler period from approximately 1300–1850 AD (not a full ice age). Glaciers advanced in Alpine Europe and North America. The Thames froze regularly. Caused by a combination of reduced solar output and increased volcanic activity. Ended as the Industrial Revolution began warming the climate.

Why Glaciers Form

A glacier forms where **snowfall exceeds snowmelt** over many years, and snow compresses into ice. Two key zones govern glacier behaviour:

Zone	Process	Location on glacier
Accumulation zone	Snow and ice build up (inputs exceed outputs). Net gain of ice mass.	Upper part of glacier / high altitude

Ablation zone	Ice is lost through melting, evaporation, and calving. Outputs exceed inputs.	Lower part of glacier / lower altitude / snout

Glacier mass balance (glacial budget): The difference between accumulation (gains) and ablation (losses) over one year. If accumulation > ablation, the glacier has a *positive mass balance* and advances. If ablation > accumulation, it has a *negative mass balance* and retreats. The **equilibrium line** (or firn line) marks where accumulation = ablation.

Exam tip: Questions often ask you to explain what a positive or negative mass balance means for glacier movement. Always link back to the balance between accumulation and ablation, and state whether the glacier advances or retreats.

2. Glacial Processes

Freeze-Thaw Weathering (Frost Shattering)

Freeze-thaw weathering: A mechanical weathering process that breaks rock apart without chemical change. Water enters cracks in rock. When temperatures drop below 0°C, the water freezes and expands by approximately 9%. This expansion exerts pressure on the rock (up to 2,000 kPa). When temperatures rise, the ice melts and pressure is released. Repeated freeze-thaw cycles gradually widen cracks until angular fragments of rock (called *scree* or *frost-shattered debris*) break off.

This is most effective where temperatures frequently oscillate around 0°C — common in periglacial and alpine environments. The angular fragments produced by freeze-thaw are important because they provide the sharp, abrasive tools embedded in glacier ice.

Plucking (Quarrying)

Plucking: A glacial erosion process in which the glacier freezes onto the bedrock. Meltwater percolates into cracks in the rock beneath the glacier and re-freezes, bonding the ice to rock fragments. As the glacier moves forward, it *pulls (plucks) these rock fragments away* from the bedrock. Most effective on the lee (downslope) side of rock outcrops. Produces a rough, jagged surface.

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Abrasion

Abrasion: A glacial erosion process in which rocks, boulders, and sediment embedded in the base and sides of the glacier act like sandpaper, grinding against the bedrock as the glacier moves. This smooths and polishes the bedrock surface, producing *striated rock pavements* (parallel scratches called striations that indicate the direction of ice movement). Finer particles produce *rock flour* (glacial milk) that gives glacial meltwater its characteristic milky appearance.

Rotational Slip in Corries

Rotational slip: The movement of ice within a corrie (cirque) in a curved, rotational manner. The ice pivots around a point, eroding the base of the hollow more deeply than the lip, creating the characteristic over-deepened basin shape of a corrie. This rotational movement is caused by the weight of accumulated ice and gravity acting on the curved corrie floor.

Exam tip: Know the difference between plucking and abrasion. Plucking produces angular, blocky material on the lee side of rock bumps; abrasion produces smooth, polished, scratched surfaces. Examiners often ask you to distinguish between them.

3. Glacial Erosion Landforms

Corries (Cwms / Cirques)

Corrie (cwm/cirque): An armchair-shaped hollow on a mountainside, formed by glacial erosion. It has a steep back wall (headwall), a deep over-deepened basin (often containing a tarn lake), and a raised lip or rock threshold at the front. The Welsh term is *cwm* (e.g. Cwm Idwal, Snowdonia); the French/Scottish term is *corrie*.

Formation: Snow accumulates in a sheltered hollow (usually north or north-east facing in the UK to avoid direct sunlight). Snow compresses into ice. Rotational slip deepens the hollow; freeze-thaw shatters the backwall; plucking removes rock from the backwall and base. The lip forms where erosion is less intense.

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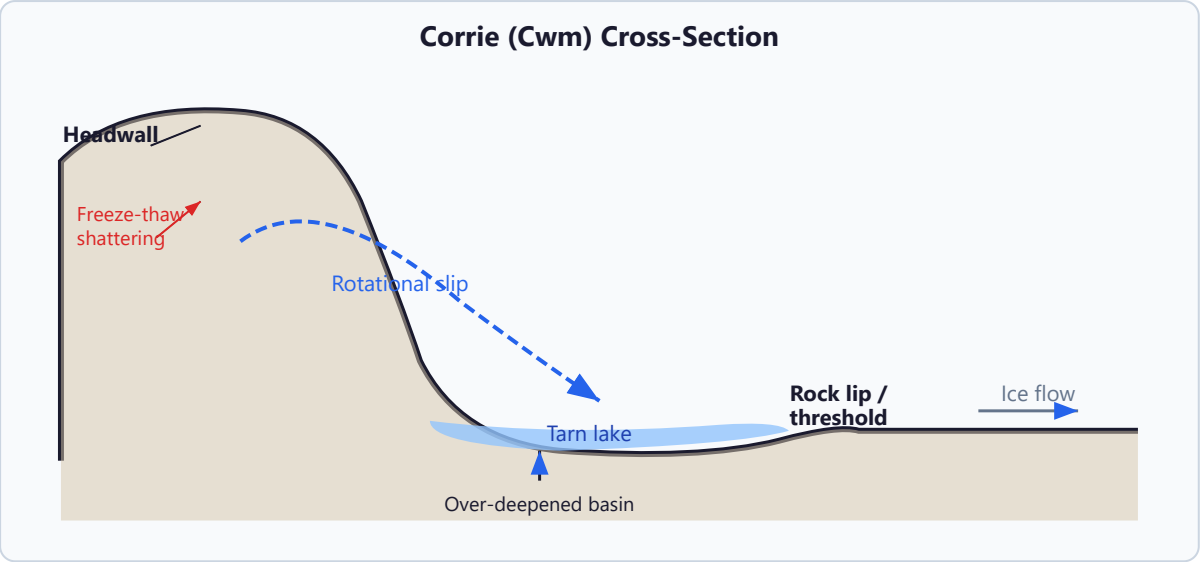


Arêtes

Arête: A narrow, knife-edged ridge formed when two corries develop back to back or side by side. Glacial erosion deepens both corries, leaving only a thin, jagged rock ridge between them. Freeze-thaw weathering further sharpens the ridge.

Named examples:

- **Striding Edge**, Lake District — a classic arête on Helvellyn, popular with walkers
- **Crib Goch**, Snowdonia — a dramatic arête forming part of the Snowdon Horseshoe ridge walk

Pyramidal Peaks

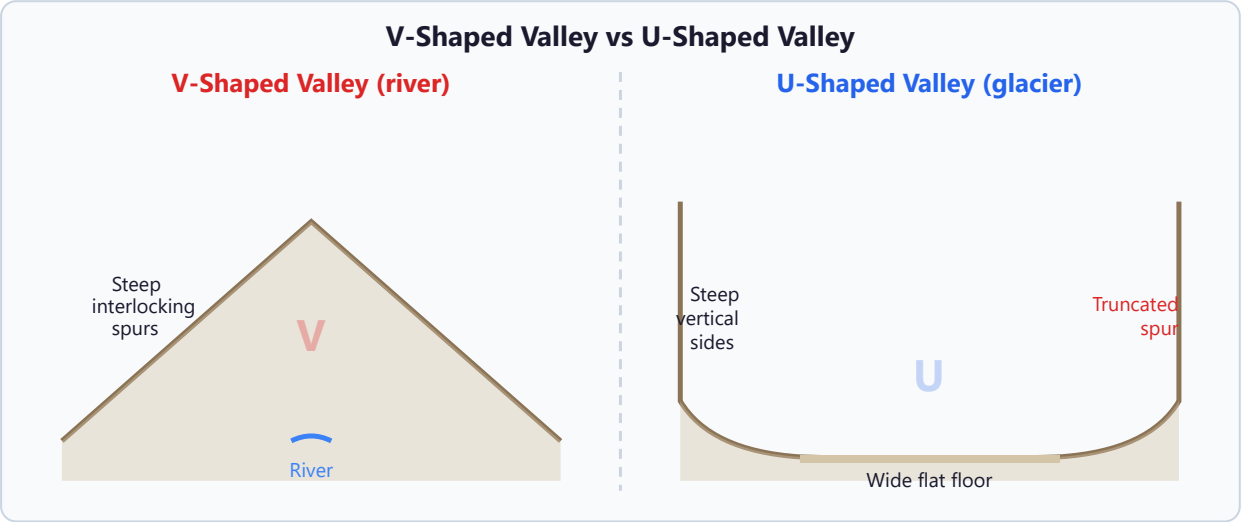
Pyramidal peak (glacial horn): A steep, angular mountain summit formed when three or more corries erode back-to-back around a central mountain. The interlocking arêtes converge at the summit, producing a pointed, pyramid-shaped peak. Freeze-thaw weathering and plucking sharpen the summit.

Named examples:

- **The Matterhorn**, Switzerland/Italy — one of the world's most iconic pyramidal peaks (4,478 m)
- **Snowdon (Yr Wyddfa)**, Wales — the highest peak in Wales, partly shaped by converging corries

U-Shaped Valleys (Glacial Troughs)

U-shaped valley (glacial trough): A wide, flat-floored, steep-sided valley formed when a glacier moves through and erodes a pre-existing river V-shaped valley. Abrasion and plucking widen and deepen the valley floor. The original interlocking spurs are cut off (truncated), leaving steep, straight valley sides. After the ice melts, the flat valley floor may be occupied by a ribbon lake or river.



Truncated Spurs

Truncated spur: A ridge of land that originally projected into a river valley as an interlocking spur. As a glacier moved through the valley, it cut through these spurs, leaving steep, blunt cliff faces (truncated ends) along the valley sides. They are a key feature of glacial troughs.

Hanging Valleys

Hanging valley: A tributary valley whose floor is at a much higher elevation than the floor of the main glacial trough below. The main glacier erodes more deeply (due to its greater volume and erosive power) than the smaller tributary glacier. After glaciation, the tributary valley is left “hanging” high above the main valley. A waterfall often cascades from the hanging valley into the trough below.

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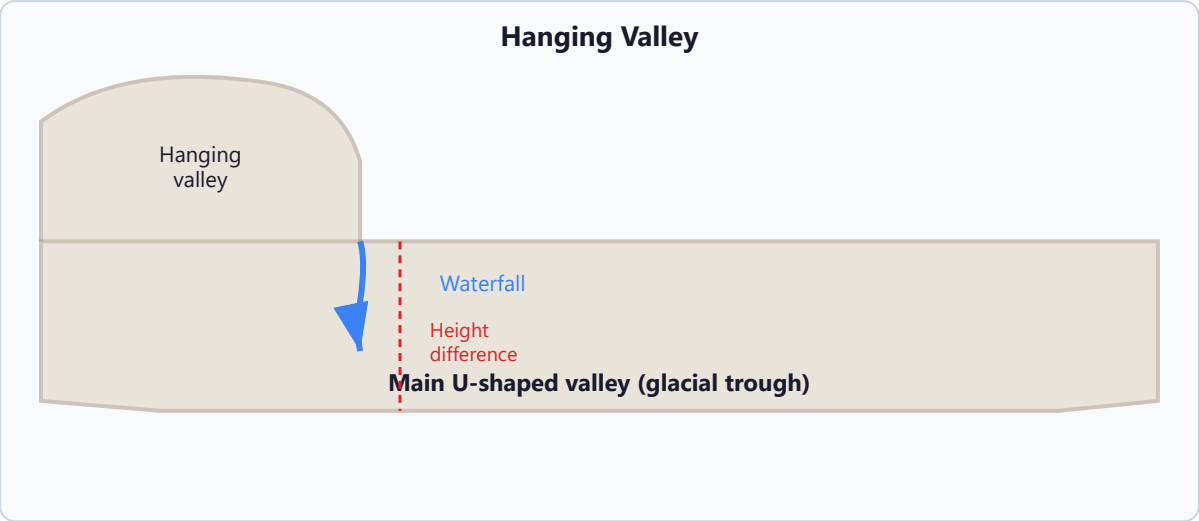


Ribbon Lakes

Ribbon lake: A long, narrow lake occupying the floor of a glacial trough. Forms where a glacier over-deepened the valley floor (often in areas of weaker rock, or where ice was particularly thick). After the ice retreats, the depression fills with water. The lake is dammed by a rock bar or terminal moraine at the lower end.

Examples: Llyn Padarn and Llyn Peris, Snowdonia; Windermere and Ullswater, Lake District; Loch Lomond, Scotland.

Exam tip: When asked to describe how a U-shaped valley forms, use a named sequence: 1) Pre-glacial V-shaped river valley; 2) Glacier forms and moves down the valley; 3) Abrasion and plucking widen and deepen the floor; 4) Interlocking spurs are cut off as truncated spurs; 5) Glacier retreats, leaving wide flat floor sometimes filled by a ribbon lake.

4. Glacial Deposition

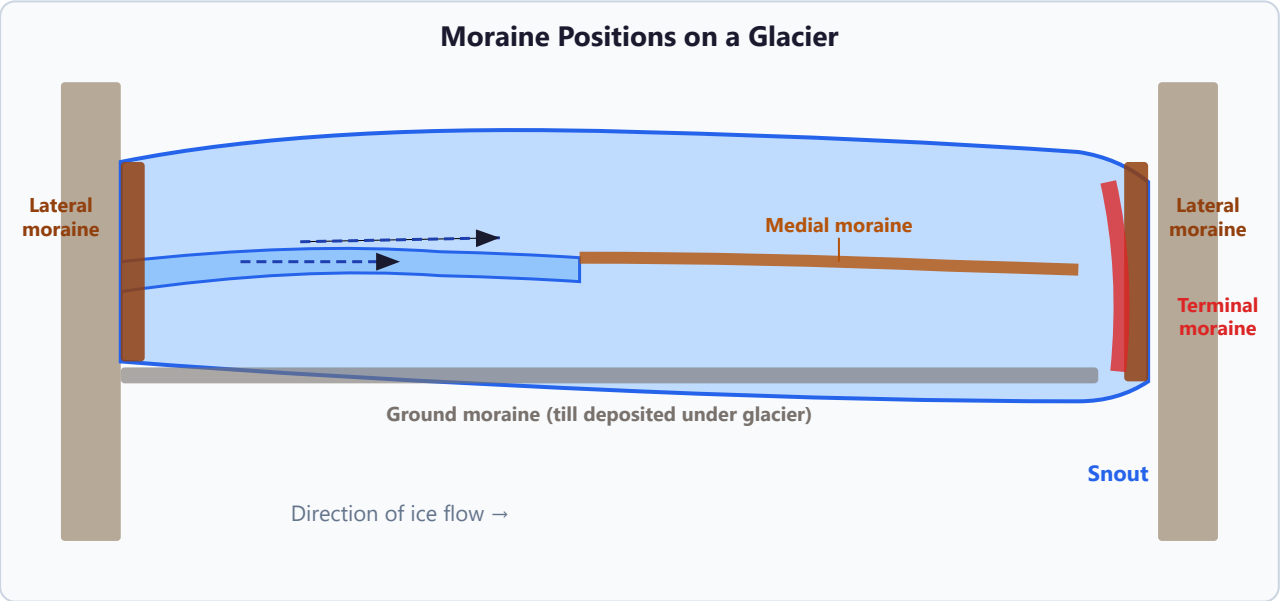
When a glacier loses energy (typically at the snout or when ice melts), it deposits the material it has been carrying. Glacial deposits are collectively called **drift**.

Types of Glacial Deposit

Type	Characteristics	Deposited by
Till (boulder clay)	Unstratified (no layers), unsorted (mixed particle sizes from clay to boulders), angular particles (not rounded by water), deposited directly by ice	Glacier ice
Fluvioglacial deposits (outwash / glaciofluvial)	Stratified (deposited in layers), sorted (particles graded by size — coarser nearer glacier, finer further away), rounded and smoothed (transported by meltwater)	Meltwater streams

Types of Moraine

Moraine: Deposits of till (rock debris) carried and deposited by glaciers. Different types of moraine form in different positions relative to the glacier.



Type of Moraine	Position	Description
Lateral moraine	Along valley sides	Debris from freeze-thaw on valley walls falls onto the glacier edges and is carried along the sides

Medial moraine	Down the centre of the glacier	Forms when two glaciers with lateral moraines merge — the two inner lateral moraines combine to form a central ridge of debris
Terminal (end) moraine	At the snout of the glacier	Marks the furthest advance of the glacier. A ridge of deposited till bulldozed to the front of the glacier. Can dam ribbon lakes after ice retreats
Ground moraine	Beneath the glacier	Till deposited across the valley floor as the glacier moves or retreats. Produces a gently undulating landscape

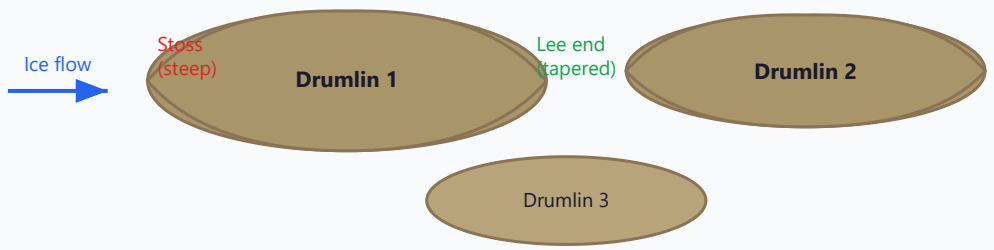
Erratics

Erratic: A rock fragment that has been transported by a glacier and deposited in an area of different geology — it does not match the local rock type. Erratics can be used to trace former glacier paths. **Example: Norber erratics**, North Yorkshire — Silurian greywacke boulders (c. 480 Ma old) transported and deposited on a limestone pavement of Carboniferous limestone (c. 340 Ma old) by a glacier from Crummackdale.

Drumlins

Drumlin: A smooth, elongated, egg-shaped hill composed of till, formed beneath a moving glacier. The *stoss end* (upstream, facing the direction the ice came from) is steep and blunt; the *lee end* (downstream) is gentle and tapered. Drumlins are usually found in groups (swarms). Their long axis is parallel to the direction of ice flow.

Drumlin Shape and Swarm Pattern (plan view)



Drumlins form in swarms ("basket of eggs" topography)
Long axis parallel to ice flow direction

Exam tip: The key distinction between till and fluvioglacial deposits is **stratification and sorting**. Till = unstratified, unsorted, angular. Fluvioglacial = stratified (layered), sorted, rounded. This tells you whether it was deposited directly by ice or by meltwater.

5. Periglaciation

Periglacial environment: Areas that are not covered by ice but experience very cold conditions, freeze-thaw cycles, and the presence of permafrost. Located at the margins of glaciers and ice sheets, or at high altitudes. Today found in Siberia, Alaska, northern Canada, Scandinavia, Tibet, and on high mountain summits.

Permafrost

Permafrost: Ground that remains frozen (below 0°C) for at least two consecutive years. The surface layer (the *active layer*) may thaw in summer. Three types:

- **Continuous permafrost** — covers entire area; temperature always well below 0°C. Found closest to poles (e.g. northern Siberia, northern Alaska). Depth can reach 1,500 m.
- **Discontinuous permafrost** — covers most but not all the ground (60–90%). Warmer areas (e.g. under lakes, rivers, forest) may be unfrozen. Found slightly further from poles.
- **Sporadic permafrost** — only isolated patches remain frozen. Occurs in the southernmost margins of permafrost zones or at high altitudes in temperate regions.

Freeze-Thaw and Solifluction

Solifluction: The slow downslope movement of saturated soil and regolith over frozen ground (permafrost). In summer, the active layer thaws and becomes waterlogged (meltwater cannot drain through the frozen layer below). Gravity causes this saturated material to flow slowly downslope, even on gentle gradients (as low as 2–3°). Produces *solifluction lobes* — tongue-shaped masses of material on hillsides.

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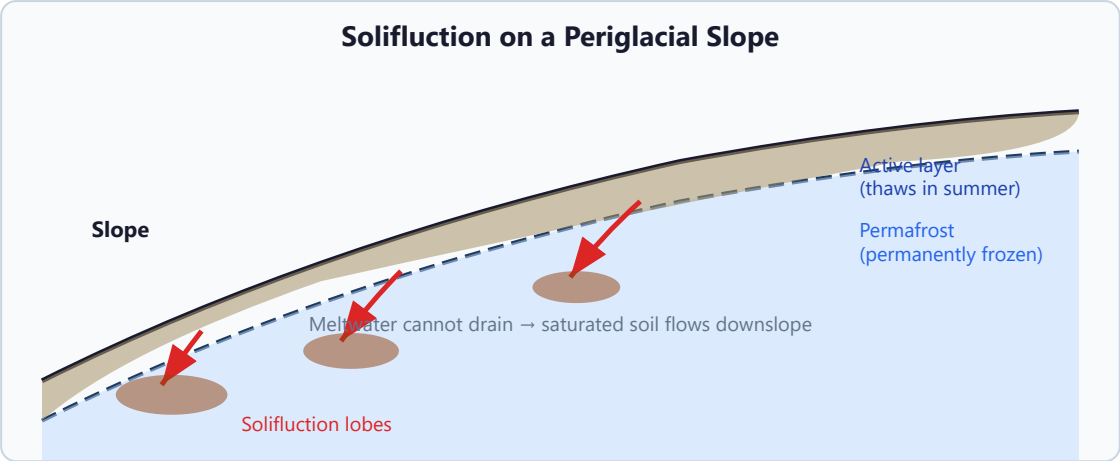

Patterned Ground

Patterned ground: Geometric patterns formed on flat or gently sloping periglacial surfaces by repeated freeze-thaw cycles. Freezing soil expands and pushes larger stones outward and upward; on flat ground this produces *stone polygons* (polygon-shaped outlines of stones); on slopes, gravity elongates these into *stone stripes* running downslope.

Pingos

Pingo: A large dome-shaped hill (up to 50 m high, 600 m wide) with an ice core, found in periglacial regions. Two types: *Open-system pingos* form on slopes where groundwater migrates upward and freezes; *closed-system pingos* form in lake beds as permafrost advances inward, forcing water upward which freezes into an ice lens, doming up the overlying sediment. When the ice melts, a circular hollow with a raised rim is left (called a *pingo ruin*).

Exam tip: Periglacial questions often appear on Paper 1. Key terms to know: permafrost, active layer, solifluction, patterned ground, and pingo. Make sure you can describe the process of solifluction step by step, linking permafrost to waterlogging to downslope movement.

6. Human Activity in Glaciated Uplands

Glaciated uplands in the UK (such as the Lake District, Snowdonia, the Cairngorms, and the Pennines) offer a range of opportunities but also present significant challenges for human use.

Activity	Example	Opportunities	Challenges
Farming	Sheep grazing on upland fells (e.g. Herdwick sheep, Lake District)	Traditional land use; wool and meat production; maintains open moorland	Thin, acidic soils; short growing season; steep gradients; harsh winters; low income
Forestry	Kielder Forest, Northumberland — UK's largest man-made forest	Timber production; carbon sequestration; employment; wildlife habitat	Non-native conifer monocultures reduce biodiversity; visual impact on landscape; acidifies soils and streams
Tourism / Recreation	Cairngorms ski resort; hiking in Snowdonia / Lake District	Major income source; employment; supports local services; footpath access	Footpath erosion; traffic congestion; litter; pressure on local housing; seasonal employment
Quarrying	Slate quarrying (e.g. Dinorwig, Snowdonia)	Raw material supply (slate, granite, aggregate); employment; local infrastructure	Landscape destruction; noise and dust pollution; heavy lorry traffic; habitat loss
Hydro-electric power (HEP)	Dinorwig Power Station ('Electric Mountain'), Snowdonia; Cruachan, Scotland; Ben Nevis area	Renewable, low-carbon electricity; pumped storage provides grid flexibility; economic benefit	Flooding valleys for reservoirs; visual impact of infrastructure; disruption to river ecosystems; high construction costs

Exam tip: For 6-mark questions on opportunities and challenges, aim to refer to at least three different land uses and use named examples. Always try to evaluate — state whether the challenges outweigh the opportunities or vice versa, and why.

7. Case Study — Snowdonia (Eryri) / Lake District

Case Study: Snowdonia National Park (Parc Cenedlaethol Eryri), Wales

Glacial Landforms Present

Landform	Named Example in Snowdonia	Description
Corrie (Cwm)	Cwm Idwal	A well-preserved glacial cwm with a tarn (Llyn Idwal). Classic headwall, rock lip, and over-deepened basin. Home to rare arctic-alpine plants.
Arête	Crib Goch	A knife-edged ridge between two corries; part of the Snowdon Horseshoe route. Grade 1 scramble.
Pyramidal peak	Snowdon (Yr Wyddfa) 1,085 m	The highest point in Wales, shaped by converging corries. Visited by c. 600,000 people/year.
U-shaped valley	Nant Ffrancon valley	Classic glacial trough; flat valley floor now occupied by the A5 road and the Afon Ogwen river
Ribbon lake	Llyn Padarn / Llyn Peris	Long, narrow lakes in the Llanberis valley occupying an over-deepened glacial trough
Hanging valley	Cwm Bochlwyd above Nant Ffrancon	Tributary cwm left high above the main trough; the outflow falls steeply to the valley floor
Truncated spur	Valley sides of Nant Ffrancon	Spur ends cut off by ice, leaving steep blunt cliff faces visible on the valley walls

Tourism: Scale and Impacts

Snowdonia receives approximately **4.2 million visitors per year**. Around 600,000 climb Snowdon alone annually.

Positive economic impacts:

- Tourism generates approximately £396 million/year for the local economy
- Supports 5,900 jobs in the national park area

- Revenue supports services such as the Snowdon Mountain Railway, cafes, and accommodation
- Attracts investment in infrastructure (car parks, visitor centres, footpaths)

Negative environmental impacts:

- Severe footpath erosion on Snowdon summit ridges — path widths expanding from 1 m to >5 m in places
- Litter and waste on summit and popular trails
- Disturbance to rare habitats: Cwm Idwal is a National Nature Reserve — trampling threatens arctic-alpine plant communities including Snowdon lily (*Gagea serotina*)
- Traffic congestion in Llanberis and Betws-y-Coed; car parks overflowing

Management Strategies

- **Footpath repair:** Snowdonia National Park Authority spends >£500,000/year on path maintenance; helicopter-airlifted stone slabs used on high-level paths
- **Warden service:** Rangers patrol popular areas to educate visitors and prevent damage
- **Cwm Idwal NNR:** Fencing to protect rare plants, interpretation boards, restricted climbing areas
- **Sustainable transport:** Snowdon Sherpa bus service to reduce car use on mountain roads
- **Leave No Trace campaigns:** Bilingual (Welsh/English) signage promoting responsible visiting
- **Visitor dispersal:** Promotion of quieter areas (e.g. Rhinog mountains) to reduce pressure on Snowdon

Additional Case Study: Lake District National Park, England

The Lake District was designated England's first **UNESCO World Heritage Site** in 2017, recognising its outstanding glacial landscape and cultural significance.

Key glacial features:

- **Ribbon lakes:** Windermere (England's longest lake, 17 km), Ullswater, Thirlmere, Coniston Water — all occupy glacially over-deepened troughs
- **U-shaped valleys:** Borrowdale, Great Langdale, Patterdale
- **Arêtes:** Striding Edge and Swirral Edge on Helvellyn
- **Pyramidal peak:** Scafell Pike (978 m) — England's highest mountain

- **Drumlins:** Eden Valley to the east of the Lake District hills contains notable drumlin swarms

Tourism: c. 18 million visitors/year (one of UK's most visited destinations). Key management challenge: Windermere traffic and watercraft causing erosion of lake shores and noise disturbance to wildlife.

Exam tip: For case study questions, always use **specific named examples** — name the corrie (Cwm Idwal), the ribbon lake (Llyn Padarn), the arête (Crib Goch). Generic answers score fewer marks. Learn at least three named landforms and three specific management strategies.

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